

Oracle Ksplice

Objectives

After completing this lesson, you should be able to:

- Describe the purpose of Oracle Ksplice
- Describe the features, benefits, and operation of Oracle Ksplice
- Implement the Ksplice Online Standard Client
- View Ksplice packages on ULN
- Describe Ksplice implementation in Oracle Cloud Infrastructure
- Use Ksplice commands
- Use the Ksplice web interface
- Use the Ksplice Offline Standard Client
- Implement the Ksplice Online Enhanced Client
- Configure Ksplice Offline Clients to use a local Ksplice mirror



Oracle Ksplice: Introduction

Ksplice:

- Updates the kernel and select user space libraries on a running system
- Applies the latest security errata (CVEs) and critical bug fixes
 - Patches are effective immediately without rebooting
- Does not halt the system
- Does not restart applications
- Applies updates in the background
- Requires an Oracle Premier support subscription
- Works with both the Unbreakable Enterprise Kernel and Red Hat Compatible Kernel
- Has an easy-to-use website



Ksplice Addresses System Administrator Challenges

- Security updates and crucial bug fixes need to be applied.
 - Results in down time
 - Maintenance windows must be coordinated and scheduled.
 - Time and disruption is costly.
- Administrators must balance keeping systems up-to-date and secure versus down time.
- Ksplice mitigates these challenges.
 - Updates are applied with no down time.
 - Updates are fast.
 - Systems are secured and bug fixes are applied quickly.
 - System availability is improved.
 - Costs are lowered.



Ksplice: Features and Benefits

Features	Benefits
• Update/patch Oracle Linux systems with no down time • Roll back updates	• Systems remain available and are secured in real time • Reduces cost/disruption • Updates can be removed without reboot • Can integrate with change control
• Web and API interfaces • No agents or daemons	• Centralized management • No impact on performance
• Oracle Linux on bare metal, Oracle VM, or Oracle Cloud supported	• Multiple platforms
• Patching supported with applications, middleware, and database installed	• No impact on running Oracle software

Ksplice Patching

- Oracle Linux
 - The Unbreakable Enterprise Kernel (UEK)
 - Red Hat Compatible Kernel (RHCK)
- Oracle VM Server (dom0)
 - VMs continue running
 - No migrations needed
- User space RPMs
 - glibc
 - OpenSSL
 - Must install Ksplice-aware versions from ULN with one-time reboot
- Scalability
 - Up to 5,000 kernel versions at a time, for widespread issues
 - Complex patches are handled

Ksplice for Diagnosing Issues

- Work with Oracle support.
 - Create a Service Request.
- If the issue has no known resolution, support can work with development for a diagnostic Ksplice patch.
 - Collect further data/information for diagnosis.
- After the resolution is identified, the Ksplice diagnostic patch is removed.
- A final Ksplice patch is developed for issue resolution.
 - Errata is provided to Unbreakable Linux Network (ULN).
 - The final Ksplice patch can be applied using uptrack.
 - It is available to Oracle Linux Premier Support customers.
- Everything is done with no rebooting and no down time.



Ksplice Track Record

- Over 1.5 million updates shipped
- Example of a kernel vulnerability addressed:
 - CVE-2016-5696 Challenge-ACK TCP hijacking
 - Ksplice patch available five days before releases from other vendors.
- Examples of user space vulnerabilities remedied:
 - CVE-2015-7547 Glibc vulnerabilities
 - Ksplice was the only option for patching without a restart.
 - CVE-2016-0800 (DROWN) OpenSSL vulnerability
 - Ksplice again was the only solution supporting patching without reboot.

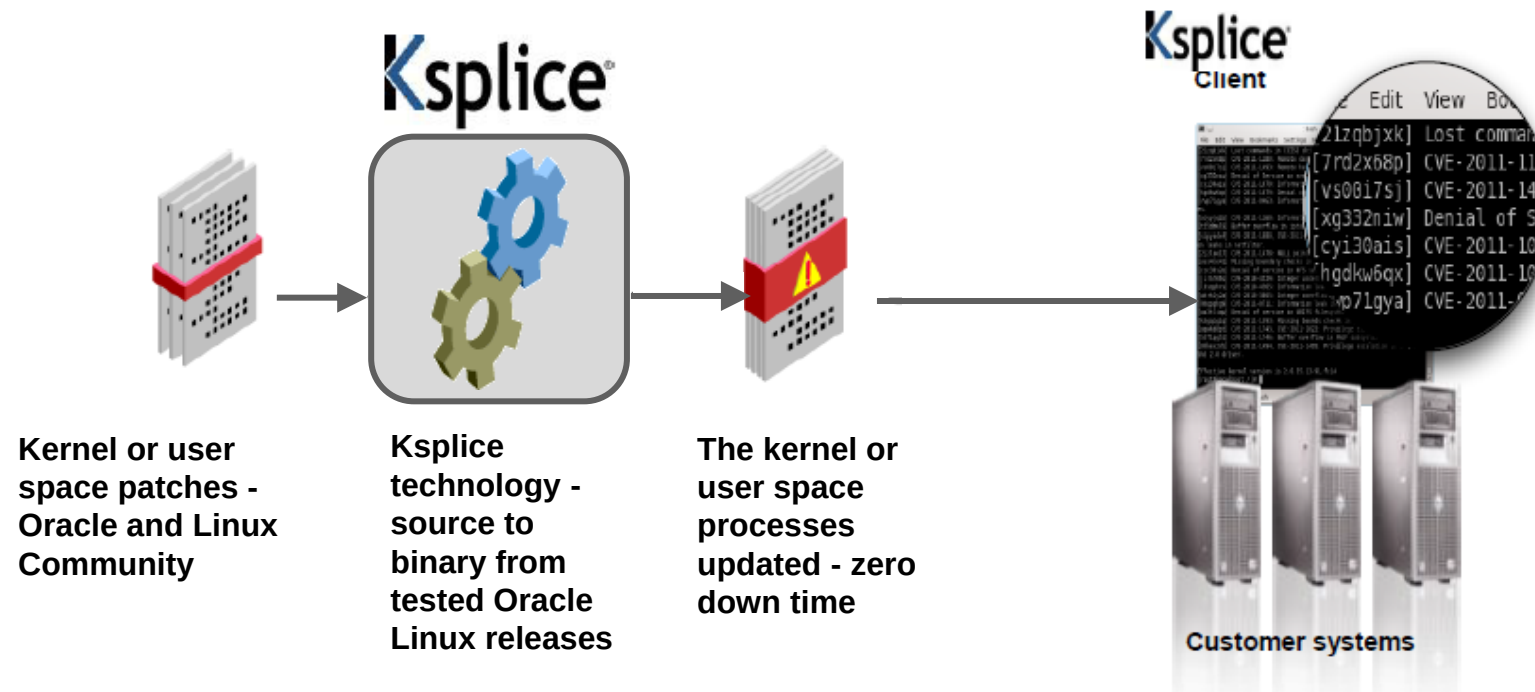


Ksplice Case Study

- Oracle Compute
 - 1,000 Oracle Linux systems running EDA simulations
 - Rolling upgrade 10% at a time
 - Each upgrade took ~54 hours of sysadmin time, 7 hours of user time lost
 - 28 days for full upgrade
- With Ksplice, time reduced to ~1 hour



How Ksplice Works



- Patches go into new kernel or user space releases, are tested by Oracle.
 - They are then turned into Ksplice updates from Oracle Linux packages.
- `glibc` and `openssl` user space libraries are updated on disk as well.
 - Updates are available as shared object files for applications to load as needed.

Online Ksplice Implementations

- For client systems with internet access
 - Installation packages are obtained from ULN.
 - Updates are obtained from the Oracle Uptrack server.
- Standard Client: Provides kernel updates
 - uptrack package installed
 - Uses Ksplice Uptrack commands and updates the kernel
- Enhanced Client: Provides kernel and user space updates
 - ksplice package installed, with dependencies (includes uptrack)
 - Ksplice-aware user space libraries, glibc and openssl, installed with one-time reboot
 - Uses Ksplice commands
 - Updates the kernel along with glibc and openssl shared libraries

Offline Ksplice Implementations

- For client systems with no internet access
 - For example: Security purposes
 - Updates are packaged in RPMs.
 - Use a local Ksplice mirror or Oracle Cloud Infrastructure mirror for offline clients to obtain installation packages and updates.
- Standard Client: Provides kernel updates
 - uptrack-offline package installed
 - Uses Ksplice Uptrack commands and updates the kernel
- Enhanced Client: Provides kernel and user space updates
 - ksplice-offline package installed, with dependencies (includes uptrack-offline)
 - Ksplice-aware user space libraries, glibc and openssl, installed with one-time reboot
 - Uses Ksplice commands
 - Updates the kernel along with glibc and openssl shared libraries

Implementing the Ksplice Online Standard Client

A summary of actions to get started with Ksplice:

1. Register your systems with Unbreakable Linux Network (ULN).
2. Subscribe to the appropriate Ksplice channel.
3. Use the `yum` command to install the `uptrack` package.
4. Perform any required configuration.
5. View the status from the System Status page of the Ksplice web interface at <https://status-ksplice.oracle.com>.

After you register your system with ULN and install the `uptrack` package, you receive an email containing instructions for logging on to the Ksplice web interface.

The Ksplice web interface can also be accessed from <https://uptrack.ksplice.com>.

Subscribing to the Ksplice Channel



Unbreakable Linux Network (ULN)

Welcome: Search Profile About SSO Logout

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[Home](#) [Systems](#) [System Detail for OracleLinux7Update5.example.com](#) [Manage Subscriptions](#)

System summary

System name **OracleLinux7Update5.example.com**
Architecture **x86_64**
OS Release **7**

Subscriptions

Channels

Available

Oracle Software for Oracle Linux 7 (x86_64)
Packages considered as preview for test and development environments for Oracle Linux 7 (x86_64)
Packages for Oracle Stack Health Checks - Oracle Linux 7 (x86_64)
Packages for creating test and development environments for Oracle Linux 7 (x86_64)
Red Hat compatible kernel with Oracle provided fixes for Oracle Linux 7 (x86_64)
Spacewalk Server 2.6 for Oracle Linux 7 (x86_64)
Spacewalk Server 2.7 for Oracle Linux 7 (x86_64)
Unbreakable Enterprise Kernel Release 3 for Oracle Linux 7 (x86_64) - Latest
Unbreakable Enterprise Kernel Release 4 for Oracle Linux 7 (x86_64) -Archive
Unbreakable Enterprise Kernel Release 5 for Oracle Linux 7 (x86_64)
Unbreakable Enterprise Kernel Release 5 for Oracle Linux 7 (x86_64) -Archive
Ksplice for Oracle Linux 7 (x86_64)

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Subscribed

Oracle Linux 7 Latest (x86_64)
Unbreakable Enterprise Kernel Release 4 for Oracle Linux 7 (x86_64)

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Ksplice Packages on ULN



Unbreakable Linux Network (ULN)

[Home](#) [Channels](#) [Systems](#) [Errata](#) [CVE](#) [CSI Administration](#)

[Home](#) > [Channels](#) > [Channel: Ksplice for Oracle Linux 7 \(x86_64\)](#) > [Packages](#)

[Channel Detail](#) [Channel Packages](#)

Channel Packages

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Package	Description
ksplice-1.0.35-1.el7.x86_64	Oracle Ksplice user-space updates
ksplice-core0-1.0.35-1.el7.x86_64	Oracle Ksplice runtime
ksplice-offline-1.0.35-1.el7.x86_64	Offline Oracle Ksplice user-space installation.
ksplice-snmp-plugin-0.1.0-1.el7.x86_64	Ksplice SNMP plugin
ksplice-tools-1.0.35-1.el7.x86_64	Oracle Ksplice management tools
python-ksplice-uptrack-0.2.3-1.el7.noarch	Python bindings for the Ksplice Uptrack API
uptrack-1.2.52-0.el7.noarch	-
uptrack-offline-1.2.52.offline-0.el7.noarch	-
uptrack-updates-3.10.0-123.1.2.el7.x86_64-20180717-0.noarch	Rebootless updates for the Ksplice Uptrack rebootless kernel update service
uptrack-updates-3.10.0-123.13.1.el7.x86_64-20180717-0.noarch	Rebootless updates for the Ksplice Uptrack rebootless kernel update service
uptrack-updates-3.10.0-123.13.2.el7.x86_64-20180717-0.noarch	Rebootless updates for the Ksplice Uptrack rebootless kernel update service
uptrack-updates-3.10.0-123.20.1.el7.x86_64-20180717-0.noarch	Rebootless updates for the Ksplice Uptrack rebootless kernel update service

Using Ksplice Uptrack

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Overview

The Uptrack suite provide tools for managing Ksplice updates on your systems. To support various environments, Ksplice provides **command line tools** and a **graphical interface**.

Command Line Tools

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uptrack-upgrade

Ksplice updates are the same security and bug-fix updates you would get from your Linux vendor, packaged in a special reboot-less form. To apply Ksplice updates, just run **uptrack-upgrade** :

```
# uptrack-upgrade -y
The following steps will be taken:
Install [guclwyc2] CVE-2012-0957: Information leak in uname syscall.
Install [j4d07e02] Kernel panic in IPv4 ARP and IPv6 Neighbor Discovery.
Install [r8og1ec4] CVE-2013-1979: Privilege escalation with UNIX socket credentials.
Install [fiq04xbb] CVE-2013-2237: Information leak on IPsec key socket.
Install [9q4luou3] CVE-2014-3687: Remote denial-of-service in SCTP stack.
Installing [guclwyc2] CVE-2012-0957: Information leak in uname syscall.
Installing [j4d07e02] Kernel panic in IPv4 ARP and IPv6 Neighbor Discovery.
Installing [r8og1ec4] CVE-2013-1979: Privilege escalation with UNIX socket credentials.
Installing [fiq04xbb] CVE-2013-2237: Information leak on IPsec key socket.
Installing [9q4luou3] CVE-2014-3687: Remote denial-of-service in SCTP stack.
Your kernel is fully up to date.
Effective kernel version is 2.6.39-400.215.13.el6uek
#
```

Hide

Uptrack User's Guide

- Overview
- Command Line Tools
 - [uptrack-upgrade](#)
 - [uptrack-show](#)
 - [uptrack-remove](#)
 - [uptrack-uname](#)
- Configuration
 - [Automatic updates](#)
 - [Ksplice and Kernel Versions](#)
 - [Firewall and Proxy Configuration](#)
- Graphical Interface
- Access Policies
 - [Setting Policy for Existing Machines](#)
 - [Using the Web Interface](#)
 - [Using the API](#)
 - [Setting Default Policy for New Machines](#)

Ksplice Uptrack Command Summary

- `uptrack-upgrade`: Apply available updates.
- `uptrack-install <Ksplice_ID>`: Install a specific update by specifying the Ksplice ID.
- `uptrack-show`: List the active updates in your running kernel.
- `uptrack-show --available`: List the available updates to be installed.
- `uptrack-remove`: Remove applied updates from the running system and return to the original kernel version.
- `uptrack-remove <Ksplice_ID>`: Remove a specific update by specifying the Ksplice ID.
- `uptrack-uname`: Display the effective kernel version based on active Oracle Ksplice updates.



Ksplice in Oracle Cloud Infrastructure

- Ksplice is available with Oracle Cloud Infrastructure instances.
- Ksplice for kernel patches (`uptrack`) is automatically installed on instances launched from Oracle Linux images on or after August 25, 2017.
 - You can run `uptrack-upgrade` immediately to apply available updates, with no further configuration.
 - Set `autoinstall = yes` in `/etc/uptrack/uptrack.conf` to apply updates automatically.
- For instances launched from custom images or for instances launched before August 25, 2017, the `uptrack` package must be installed.
 - If you register with ULN, the `uptrack` package can be installed from ULN with yum using the `ol7_x86_64_ksplice` channel.
- Ksplice for online user space library updates requires registration with ULN or use of a ULN mirror, with installation/setup necessary.
 - Ksplice for offline user space updates uses an Oracle Cloud Infrastructure mirror, with no ULN registration or access.

Installing uptrack for Ksplice Without ULN Registration

- If uptrack must be installed in Oracle Cloud Infrastructure and you do not register with ULN, follow these steps:
 - Set up a proxy if needed.
 - Get the `install-uptrack-oc` script:

```
# wget -N https://www.ksplice.com/uptrack/install-uptrack-oc
```

- Run the `install-uptrack-oc` script. The `--autoinstall` option causes updates to be applied automatically:

```
# sh install-uptrack-oc --autoinstall
```

- Or, the following requires manual updates:

```
# sh install-uptrack-oc
```

Ksplice Web Interface: System Status

[System Status](#) [Group Management](#) [Allow/Deny Policies](#) [Settings](#) [Feedback and Support](#)

[Active Installations](#) | [Inactive Machines](#)

Overview

Access key:

1 active machine is up to date.

1 active machine is out of date.

1 machine has [stopped using the Uptrack service.](#)

Active Installations

Show group: All machines Go [Edit groups](#)

Group	Machine	Status	Auto install	Kernel product	Original Kernel	Effective Kernel	Uptrack version
	oraclelinux7u5-min.example.com (10.0.2.25)	0 installed, 117 more	No	Oracle Linux 7 with UEK 4	4.1.12-112.16.4.el7uek	4.1.12-112.16.4.el7uek	1.2.52
	oraclelinux7update5.example.com (10.0.2.19)	Up to date! (117 installed)	No	Oracle Linux 7 with UEK 4	4.1.12-112.16.4.el7uek	4.1.12-124.19.6.el7uek	1.2.52

To install Ksplice Uptrack on more systems, please see the [Installation instructions.](#)

To remove Ksplice Uptrack from a system, please see the [Removal instructions.](#)

Inactive Machines

These systems have stopped connecting to the Ksplice Uptrack service. This might be because you removed the Uptrack software, decommissioned or reinstalled the system, or because the Uptrack software can no longer reach the Uptrack servers. This list is provided for your convenience only: **Your account will not be charged for an inactive machine after the last billing cycle in which that machine was active.**

Click 'remove' to remove a machine from this list, or [remove all](#) of these machines. If a removed machine checks in with the Uptrack server, it will reappear on the status page.

	Group	Machine	Status	Last reported	Kernel product	Original Kernel	Effective Kernel	Uptrack version
Remove		oraclelinux7update4.example.com (10.0.2.12)	Inactive, Out of date	2018-10-01 21:30:41	Oracle Linux 7 with UEK 4	4.1.12-94.3.9.el7uek.x86_64	4.1.12-94.3.9.el7uek.x86_64	1.2.52

Ksplice Web Interface: System Status Detail

The screenshot displays the Ksplice Web Interface for system status. The interface is divided into two main sections: Overview and Updates.

Overview

Hostname: oraclelinux7update5.example.com
IP address: 10.0.2.19
Last reported: 2018-10-01 20:47 (0 days, 1 hour, 0 minutes, 39 seconds ago)
Uptrack version: 1.2.52
Autoinstall: No
Original Kernel: 4.1.12-112.16.4.el7uek
Linux (x86_64) 4.1.12-112.16.4.el7uek.x86_64 #2 SMP Mon Mar 12 23:57:12 PDT 2018
Effective Kernel: 4.1.12-124.19.6.el7uek
Linux (x86_64) 4.1.12-124.19.6.el7uek.x86_64 #2 SMP Thu Sep 27 22:10:16 PDT 2018
Distribution: Oracle Linux 7 with UEK 4
Uptime: 0.02 days
Status: Your kernel is up to date.
Group: [\(edit groups\)](#)
[No group]

This machine is currently **allowed** to receive updates.

View list of machines

Available Updates
No updates are available.

Installed Updates

- ✓ q0j0yb6c KAISER/KPTI enablement for Ksplice.
- ✓ afoeymft Improve the interface to freeze tasks.
- ✓ bohqh05m CVE-2017-17052: Denial-of-service due to incorrect reference counting in fork.
- ✓ eo2kqthd Weakness when checking the keys in the XTS crypto algorithm.
- ✓ nq1xhhj5 CVE-2018-7492: Denial-of-service when setting options for RDS over Infiniband socket.
- ✓ b1gg8wsq CVE-2017-7518: Privilege escalation in KVM emulation subsystem.

The Ksplice Offline Standard Client

- Systems running the Ksplice Offline Standard Client do not need a network connection to the Oracle Uptrack server.
- Ksplice updates for each supported kernel version are bundled into an RPM and made available from ULN.
 - Ksplice updates for Oracle Linux 7 x86_64 systems are available on the `ol7_x86_64_ksplice` ULN channel.
- Download the RPM from ULN to a memory stick and then use the `rpm` command to install the update package.
- Alternatively, create a Local Yum Server that acts as a Ksplice mirror.
 - A Local Yum Server requires a network connection to ULN.
 - Ksplice Offline Standard Clients require access only to the Local Yum Server.
- In Oracle Cloud Infrastructure, an Oracle Cloud Infrastructure mirror is used for offline updates.

Modifying a Local Yum Server to Act as a Ksplice Mirror

- A Local Yum Server must be registered with ULN.
- Perform the configuration steps as follows:
 1. Log in to ULN at <http://linux.oracle.com>.
 2. From the ULN Systems tab, click the link for your system in the list of registered machines.
 3. On the System Details page, click Edit.
 4. On the Edit System Properties page, verify that the Yum Server check box is selected and click Apply Changes.
 5. On the System Details page, click Manage Subscriptions.
 6. On the System Summary page, select the needed channels.
 7. When you have finished selecting channels, click Save Subscriptions and log out of ULN.

Updating a Local Yum Server with Ksplice Channels

- After Ksplice channels have been added to the local yum server, repositories are updated daily by the `/usr/sbin/uln-yum-mirror` script.
- The `uln-yum-mirror` package is installed when creating a local ULN mirror and is a prerequisite to mirroring Ksplice channels.
- This package installation adds an `anacron` job that runs once per day:
 - `/etc/cron.daily/uln-yum-mirror`
- To synchronize the local Ksplice channels without waiting for the `anacron` job to run, or if the `anacron` job has been disabled, run the `uln-yum-mirror` script manually:

```
# /usr/bin/uln-yum-mirror
```


Configuring Ksplice Offline Standard Clients to Use the Local Ksplice Mirror

To configure a system as a Ksplice Offline Standard Client, perform the following steps:

1. In the `/etc/yum.repos.d` directory, edit any existing repository files and disable all entries by setting `enabled=0`.
2. In the `/etc/yum.repos.d` directory, create the `local-yum.repo` file.
3. Install the Ksplice Offline Standard Client with the following command:

```
# yum install uptrack-offline
```

4. Install the Ksplice updates that are available for the kernel:

```
# yum install uptrack-updates-`uname -r`
```

The command given above installs the Ksplice updates and applies them. It is recommended that you set up a daily `cron` job to perform this task.

The Ksplice Enhanced Client

- Is available for Oracle Linux 6 and Oracle Linux 7
- Allows updates to Ksplice-aware `glibc` and `openssl` shared libraries, in addition to kernel updates
- Online and offline configurations can be set up.
- Managed with the `ksplice` command instead of `uptrack` commands

Implementing the Ksplice Online Enhanced Client

1. Register your systems with the Unbreakable Linux Network (ULN) or have access to ULN channels on a mirror.
2. Subscribe to the appropriate Ksplice channels.
3. Use the `yum` command to install the `ksplice` package.
4. Install the Ksplice-aware versions of the `glibc` and `openssl` user space libraries.
5. Perform any required configuration.
6. Reboot your system once to use the new user space libraries.
7. Install available user space and kernel updates.
8. View the status from the System Status page of the Ksplice web interface at <https://status-ksplice.oracle.com>.

The Ksplice web interface can also be accessed from <https://uptrack.ksplice.com>.

Managing the Ksplice Enhanced Client

- Use the `ksplice` command instead of `uptrack` commands. The following are examples:
 - Display user space processes the client can patch:
`# ksplice all list-targets`
 - Display updates that have been applied:
`# ksplice all show`
 - List available kernel updates:
`# ksplice -n kernel upgrade`
 - Install available user space updates:
`# ksplice -y user upgrade`
 - Show the effective kernel:
`# ksplice kernel uname -r`
- See the `ksplice(8)` man page for further details.

Configuring Ksplice Offline Enhanced Clients to Use a Local Ksplice Mirror

After configuring a local Ksplice mirror:

1. In the `/etc/yum.repos.d` directory, edit any existing repository files and disable all entries by setting `enabled=0`, and then create the `local-yum.repo` file.
2. Install the Ksplice Offline Enhanced Client:

```
# yum install ksplice-offline
```
3. Install the following user space packages:

```
# yum install ksplice-updates-glibc ksplice-updates-openssl
```
4. Install the kernel package and apply updates to your kernel version:

```
# yum install uptrack-updates-`uname -r`
```
5. Install updates to Ksplice-aware user space libraries:

```
# yum update glibc* openssl*
```
6. Reboot once to make the Ksplice-aware user space libraries available.

Quiz



Ksplice kernel patches are effective immediately without requiring a reboot.

- a. True
- b. False



Quiz



Which of the following commands is primarily used to manage the Ksplice Enhanced Client?

- a. `uptrack`
- b. `ksplice`



Summary

In this lesson, you should have learned how to:

- Describe the operation and features of Oracle Ksplice
- Implement the Ksplice Online Standard Client
- View Ksplice packages on ULN
- Describe Ksplice implementation in Oracle Cloud Infrastructure
- Use Ksplice commands
- Use the Ksplice web interface
- Use the Ksplice Offline Standard Client
- Implement the Ksplice Online Enhanced Client
- Configure Ksplice Offline Clients to use a local Ksplice mirror



Practice 10: Overview

This practice covers the following topics:

- Using Ksplice Uptrack
- Installing the Ksplice Offline Client and kernel updates